6.7 Singular Value Decomposition (SVD)

The Singular Value Decomposition is a highlight of linear algebra. A is any m by n matrix, square or rectangular. Its rank is r. We will diagonalize this A, but not by $S^{-1}AS$. The eigenvectors in S have three big problems: They are usually not orthogonal, there are not always enough eigenvectors, and $Ax = \lambda x$ requires A to be square. The ***singular vectors*** of A solve all those problems in a perfect way.

The price we pay is to have two sets of singular vectors, $\boldsymbol{u}$'s and $\boldsymbol{v}$'s. The $\boldsymbol{u}$'s are eigenvectors of AA^{T} and the $\boldsymbol{v}$'s are eigenvectors of $A^{\mathrm{T}}A$. Since those matrices are both symmetric, their eigenvectors can be chosen orthonormal. In equation (13) below, the simple fact that A times $A^{\mathrm{T}}A$ is the same as AA^{T} times A will lead to a remarkable property of these $\boldsymbol{u}$'s and $\boldsymbol{v}$'s:

"A is diagonalized" $$Av_1 = \sigma_1 u_1 \quad Av_2 = \sigma_2 u_2 \quad \dots \quad Av_r = \sigma_r u_r \tag{1}$$

The singular vectors $v_1, \dots, v_r$ are in the *row space* of A. The outputs $u_1, \dots, u_r$ are in the *column space* of A. The *singular values* $\sigma_1, \dots, \sigma_r$ are all positive numbers. When the $\boldsymbol{v}$'s and $\boldsymbol{u}$'s go into the columns of V and U, orthogonality gives $V^{\mathrm{T}}V = I$ and $U^{\mathrm{T}}U = I$. The σ's go into a diagonal matrix Σ.

Just as $Ax_i = \lambda_i x_i$ led to the diagonalization $AS = S\Lambda$, the equations $Av_i = \sigma_i u_i$ tell us column by column that $\boldsymbol{AV = U\Sigma}$:

$$\begin{matrix}(m \text{ by } n)(n \text{ by } r)\\ \text{equals}\\ (m \text{ by } r)(r \text{ by } r)\end{matrix} \qquad A\begin{bmatrix} & & \\ v_1 & \cdots & v_r \\ & & \end{bmatrix} = \begin{bmatrix} & & \\ u_1 & \cdots & u_r \\ & & \end{bmatrix}\begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{bmatrix}. \tag{2}$$

This is the heart of the SVD, but there is more. Those $\boldsymbol{v}$'s and $\boldsymbol{u}$'s account for the row space and column space of A. We need $n - r$ more $\boldsymbol{v}$'s and $m - r$ more $\boldsymbol{u}$'s, from the nullspace $N(A)$ and the left nullspace $N(A^{\mathrm{T}})$. They can be orthonormal bases for those two nullspaces (and then automatically orthogonal to the first r $\boldsymbol{v}$'s and $\boldsymbol{u}$'s). Include all the $\boldsymbol{v}$'s and $\boldsymbol{u}$'s in V and U, so these matrices become *square*. ***We still have $AV = U\Sigma$.***

$$\begin{matrix}(m \text{ by } n)(n \text{ by } n)\\ \text{equals}\\ (m \text{ by } m)(m \text{ by } n)\end{matrix} \qquad A\begin{bmatrix} & & & \\ v_1 & \cdots & v_r & \cdots & v_n \\ & & & & \end{bmatrix} = \begin{bmatrix} & & & \\ u_1 & \cdots & u_r & \cdots & u_m \\ & & & & \end{bmatrix}\begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \\ & & \end{bmatrix} \tag{3}$$

The new Σ is m by n. It is just the old r by r matrix (call that Σ_r) with $m - r$ new zero rows and $n - r$ new zero columns. The real change is in the shapes of U and V and Σ. Still $V^{\mathrm{T}}V = I$ and $U^{\mathrm{T}}U = I$, with sizes n and m.

V is now a square orthogonal matrix, with inverse $V^{-1} = V^{\mathrm{T}}$. So $AV = U\Sigma$ can become $\boldsymbol{A = U\Sigma V^{\mathrm{T}}}$. This is the ***Singular Value Decomposition***:

SVD $$A = U\Sigma V^{\mathrm{T}} = u_1\sigma_1 v_1^{\mathrm{T}} + \cdots + u_r\sigma_r v_r^{\mathrm{T}}. \tag{4}$$

I would write the earlier "reduced SVD" from equation (2) as $A = U_r \Sigma_r V_r^T$. That is equally true, without the extra zeros in Σ. This reduced SVD gives the same splitting of A into a sum of r matrices, each of rank one.

We will see that $\sigma_i^2 = \lambda_i$ is an eigenvalue of $A^T A$ and also AA^T. When we put the singular values in descending order, $\sigma_1 \geq \sigma_2 \geq \ldots \sigma_r > 0$, the splitting in equation (4) gives the r rank-one pieces of A ***in order of importance.***

Example 1 When is $U\Sigma V^T$ (singular values) the same as $S\Lambda S^{-1}$ (eigenvalues)?

Solution We need orthonormal eigenvectors in $S = U$. We need nonnegative eigenvalues in $\Lambda = \Sigma$. So A must be a ***positive semidefinite (or definite) symmetric matrix*** $Q\Lambda Q^T$.

Example 2 If $A = xy^T$ with unit vectors x and y, what is the SVD of A?

Solution The reduced SVD in (2) is exactly xy^T, with rank $r = 1$. It has $u_1 = x$ and $v_1 = y$ and $\sigma_1 = 1$. For the full SVD, complete $u_1 = x$ to an orthonormal basis of u's, and complete $v_1 = y$ to an orthonormal basis of v's. No new σ's.

I will describe an application before proving that $Av_i = \sigma_i u_i$. This key equation gave the diagonalizations (2) and (3) and (4) of the SVD: $A = U\Sigma V^T$.

Image Compression

Unusually, I am going to stop the theory and describe applications. This is the century of data, and often that data is stored in a matrix. *A digital image is really a matrix of pixel values.* Each little picture element or "pixel" has a gray scale number between black and white (it has three numbers for a color picture). The picture might have $512 = 2^9$ pixels in each row and $256 = 2^8$ pixels down each column. We have a 256 by 512 pixel matrix with 2^{17} entries! To store one picture, the computer has no problem. But a CT or MR scan produces an image at every cross section—a ton of data. If the pictures are frames in a movie, 30 frames a second means 108,000 images per hour. Compression is especially needed for high definition digital TV, or the equipment could not keep up in real time.

What is compression? We want to replace those 2^{17} matrix entries by a smaller number, *without losing picture quality.* A simple way would be to use larger pixels—replace groups of four pixels by their average value. This is 4 : 1 compression. But if we carry it further, like 16 : 1, our image becomes "blocky". We want to replace the mn entries by a smaller number, in a way that the human visual system won't notice.

Compression is a billion dollar problem and everyone has ideas. Later in this book I will describe Fourier transforms (used in **jpeg**) and wavelets (now in **JPEG2000**). Here we try an SVD approach: *Replace the 256 by 512 pixel matrix by* ***a matrix of rank one:*** *a column times a row.* If this is successful, the storage requirement becomes 256 + 512 (add instead of multiply). The compression ratio (256)(512)/(256 + 512) is better than 170 to 1. This is more than we hope for. We may actually use five matrices of rank one (so a matrix approximation of rank 5). The compression is still 34 : 1 and the crucial question is the picture quality.

Where does the SVD come in? *The best rank one approximation to A is the matrix* $\sigma_1 u_1 v_1^T$. It uses the largest singular value σ_1. The best rank 5 approximation includes also $\sigma_2 u_2 v_2^T + \cdots + \sigma_5 u_5 v_5^T$. **The SVD puts the pieces of A in descending order.**

A library compresses a different matrix. The rows correspond to key words. Columns correspond to titles in the library. The entry in this *word-title matrix* is $a_{ij} = 1$ if word i is in title j (otherwise $a_{ij} = 0$). We normalize the columns so long titles don't get an advantage. We might use a table of contents or an abstract. (Other books might share the title "*Introduction to Linear Algebra*".) Instead of $a_{ij} = 1$, the entries of A can include the *frequency* of the search words. See Section 8.6 for the SVD in statistics.

Once the indexing matrix is created, the search is a linear algebra problem. This giant matrix has to be compressed. The SVD approach gives an optimal low rank approximation, better for library matrices than for natural images. There is an ever-present tradeoff in the cost to compute the u's and v's. We still need a better way (with sparse matrices).

The Bases and the SVD

Start with a 2 by 2 matrix. Let its rank be $r = 2$, so A is invertible. We want v_1 and v_2 to be perpendicular unit vectors. ***We also want Av_1 and Av_2 to be perpendicular.*** (This is the tricky part. It is what makes the bases special.) Then the unit vectors $u_1 = Av_1/\|Av_1\|$ and $u_2 = Av_2/\|Av_2\|$ will be orthonormal. Here is a specific example:

Unsymmetric matrix
$$A = \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix}. \tag{5}$$

No orthogonal matrix Q will make $Q^{-1}AQ$ diagonal. We need $U^{-1}AV$. The two bases will be different—one basis cannot do it. The output is $Av_1 = \sigma_1 u_1$ when the input is v_1. The "singular values" σ_1 and σ_2 are the lengths $\|Av_1\|$ and $\|Av_2\|$.

$$\begin{matrix} AV = U\Sigma \\ A = U\Sigma V^T \end{matrix} \qquad A\begin{bmatrix} v_1 & v_2 \end{bmatrix} = \begin{bmatrix} \sigma_1 u_1 & \sigma_2 u_2 \end{bmatrix} = \begin{bmatrix} u_1 & u_2 \end{bmatrix}\begin{bmatrix} \sigma_1 & \\ & \sigma_2 \end{bmatrix}. \tag{6}$$

There is a neat way to remove U and see V by itself. *Multiply A^T times A.*

$$A^TA = (U\Sigma V^T)^T(U\Sigma V^T) = V\Sigma^T\Sigma V^T. \tag{7}$$

U^TU disappears because it equals I. (We require $u_1^Tu_1 = 1 = u_2^Tu_2$ and $u_1^Tu_2 = 0$.) Multiplying those diagonal Σ^T and Σ gives σ_1^2 and σ_2^2. That leaves an ordinary diagonalization of the crucial symmetric matrix A^TA, whose eigenvalues are σ_1^2 and σ_2^2:

$$\begin{matrix} \textbf{Eigenvalues } \sigma_1^2, \sigma_2^2 \\ \textbf{Eigenvectors } v_1, v_2 \end{matrix} \qquad A^TA = V\begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix}V^T. \tag{8}$$

This is exactly like $A = Q\Lambda Q^T$. But the symmetric matrix is not A itself. Now the symmetric matrix is A^TA! *And the columns of V are the eigenvectors of A^TA.* Last is U:

Compute the eigenvectors v and eigenvalues σ^2 of A^TA. Then each $u = Av/\sigma$.

For large matrices LAPACK finds a special way to avoid multiplying A^TA in **svd** (A).

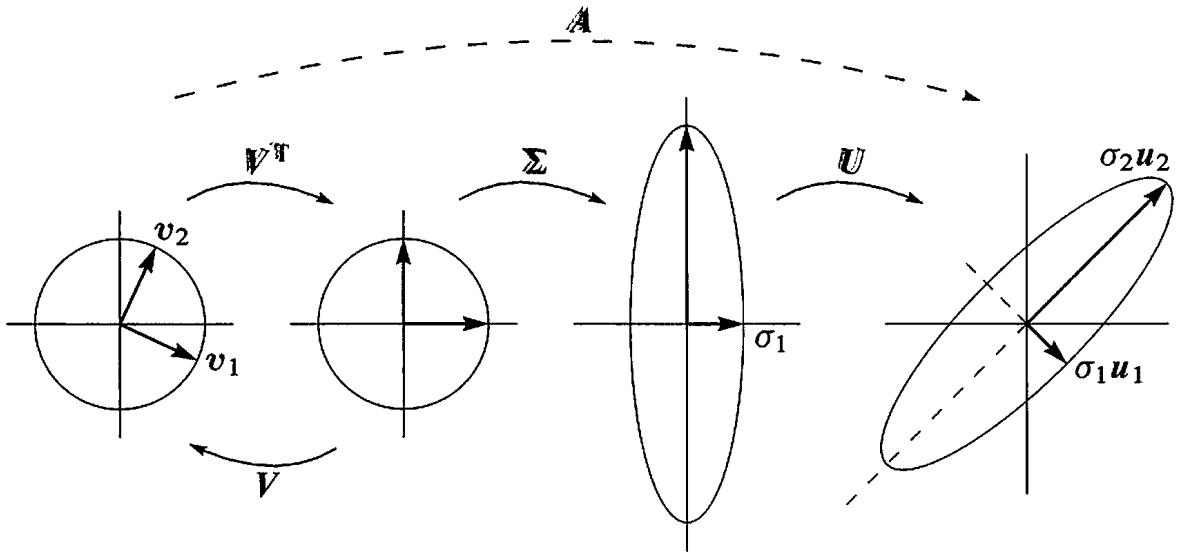


Figure 6.8: U and V are rotations and reflections. Σ stretches circle to ellipse.

Example 3 Find the singular value decomposition of that matrix $A = \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix}$.

Solution Compute A^TA and its eigenvectors. Then make them unit vectors:

$$A^TA = \begin{bmatrix} 5 & 3 \\ 3 & 5 \end{bmatrix} \quad \text{has unit eigenvectors} \quad v_1 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \quad \text{and} \quad v_2 = \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}.$$

The eigenvalues of A^TA are 8 and 2. The v's are perpendicular, because eigenvectors of every symmetric matrix are perpendicular—and A^TA is automatically symmetric.

Now the u's are quick to find, because Av_1 is going to be in the direction of u_1:

$$Av_1 = \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} = \begin{bmatrix} 2\sqrt{2} \\ 0 \end{bmatrix}. \quad \text{The unit vector is} \quad u_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Clearly Av_1 is the same as $2\sqrt{2}\,u_1$. The first singular value is $\sigma_1 = 2\sqrt{2}$. Then $\sigma_1^2 = 8$.

$$Av_2 = \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} = \begin{bmatrix} 0 \\ \sqrt{2} \end{bmatrix}. \quad \text{The unit vector is} \quad u_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Now Av_2 is $\sqrt{2}\,u_2$ and $\sigma_2 = \sqrt{2}$. Thus σ_2^2 agrees with the other eigenvalue 2 of A^TA.

$$A = U\Sigma V^T \quad \text{is} \quad \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2\sqrt{2} & \\ & \sqrt{2} \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}. \quad (9)$$

This matrix, and every invertible 2 by 2 matrix, ***transforms the unit circle to an ellipse.*** You can see that in the figure, which was created by Cliff Long and Tom Hern.

One final point about that example. We found the $\boldsymbol{u}$'s from the $\boldsymbol{v}$'s. Could we find the $\boldsymbol{u}$'s directly? *Yes*, by multiplying AA^{T} instead of $A^{\mathrm{T}}A$:

$$\text{Use } V^{\mathrm{T}}V = I \qquad AA^{\mathrm{T}} = (U\Sigma V^{\mathrm{T}})(V\Sigma^{\mathrm{T}}U^{\mathrm{T}}) = U\Sigma\Sigma^{\mathrm{T}}U^{\mathrm{T}}. \tag{10}$$

Multiplying $\Sigma\Sigma^{\mathrm{T}}$ gives σ_1^2 and σ_2^2 as before. *The $\boldsymbol{u}$'s are eigenvectors of AA^{T}*:

Diagonal in this example $$AA^{\mathrm{T}} = \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 2 & -1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 8 & 0 \\ 0 & 2 \end{bmatrix}.$$

The eigenvectors $(1,0)$ and $(0,1)$ agree with $\boldsymbol{u}_1$ and $\boldsymbol{u}_2$ found earlier. Why take the first eigenvector to be $(1,0)$ instead of $(-1,0)$ or $(0,1)$? Because we have to follow Av_1 (I missed that in my video lecture ...). Notice that AA^{T} has the same eigenvalues (8 and 2) as $A^{\mathrm{T}}A$. The singular values are $\sqrt{8}$ and $\sqrt{2}$.

Example 4 Find the SVD of the singular matrix $A = \begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix}$. The rank is $r = 1$.

Solution The row space has only one basis vector $v_1 = (1,1)/\sqrt{2}$. The column space has only one basis vector $\boldsymbol{u}_1 = (2,1)/\sqrt{5}$. Then $Av_1 = (4,2)/\sqrt{2}$ must equal $\sigma_1\boldsymbol{u}_1$. It does, with $\sigma_1 = \sqrt{10}$.

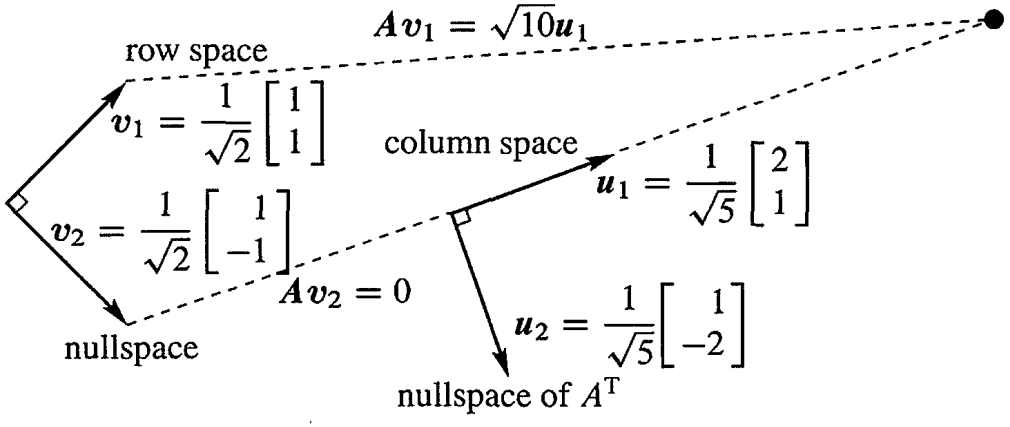


Figure 6.9: The SVD chooses orthonormal bases for 4 subspaces so that $Av_i = \sigma_i\boldsymbol{u}_i$.

The SVD could stop after the row space and column space (it usually doesn't). It is customary for U and V to be square. The matrices need a second column. ***The vector $\boldsymbol{v}_2$ is in the nullspace.*** It is perpendicular to v_1 in the row space. Multiply by A to get $Av_2 = \mathbf{0}$. We could say that the second singular value is $\sigma_2 = 0$, but singular values are like pivots—only the r nonzeros are counted.

$$\begin{matrix} A = U\Sigma V^{\mathrm{T}} \\ \textbf{Full size} \end{matrix} \qquad \begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix} = \frac{1}{\sqrt{5}}\begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix}\begin{bmatrix} \sqrt{10} & 0 \\ 0 & 0 \end{bmatrix}\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}. \tag{11}$$

The matrices U and V contain orthonormal bases for all four subspaces:

first	r	columns of V :	*row space* of A
last	$n-r$	columns of V :	*nullspace* of A
first	r	columns of U :	*column space* of A
last	$m-r$	columns of U :	*nullspace* of A^{T}

The first columns $v_1, \ldots, v_r$ and $u_1, \ldots, u_r$ are eigenvectors of $A^{\mathrm{T}}A$ and AA^{T}. We now explain why Av_i falls in the direction of u_i. The last v's and u's (in the nullspaces) are easier. As long as those are orthonormal, the SVD will be correct.

Proof of the SVD: Start from $A^{\mathrm{T}}Av_i = \sigma_i^2 v_i$, which gives the v's and σ's. Multiplying by v_i^{T} leads to $\|Av_i\|^2$. To prove that $Av_i = \sigma_i u_i$, the key step is to multiply by A:

$$v_i^{\mathrm{T}}A^{\mathrm{T}}Av_i = \sigma_i^2 v_i^{\mathrm{T}}v_i \quad \textbf{gives} \quad \|Av_i\|^2 = \sigma_i^2 \quad \textbf{so that} \quad \|Av_i\| = \sigma_i \tag{12}$$

$$AA^{\mathrm{T}}Av_i = \sigma_i^2 Av_i \quad \textbf{gives} \quad u_i = Av_i/\sigma_i \quad \textbf{as a unit eigenvector of } AA^{\mathrm{T}}. \tag{13}$$

Equation (12) used the small trick of placing parentheses in $(v_i^{\mathrm{T}}A^{\mathrm{T}})(Av_i) = \|Av_i\|^2$. Equation (13) placed the all-important parentheses in $(AA^{\mathrm{T}})(Av_i)$. This shows that Av_i is an eigenvector of AA^{T}. Divide by its length σ_i to get the unit vector $u_i = Av_i/\sigma_i$. These u's are orthogonal because $(Av_i)^{\mathrm{T}}(Av_j) = v_i^{\mathrm{T}}(A^{\mathrm{T}}Av_j) = v_i^{\mathrm{T}}(\sigma_j^2 v_j) = 0$.

I will give my opinion directly. The SVD is the climax of this linear algebra course. I think of it as the final step in the Fundamental Theorem. First come the *dimensions* of the four subspaces. Then their *orthogonality*. Then *the orthonormal bases diagonalize* A. It is all in the formula $A = U\Sigma V^{\mathrm{T}}$. You have made it to the top.

Eigshow (Part 2)

Section 6.1 described the MATLAB demo called **eigshow**. The first option is *eig*, when x moves in a circle and Ax follows on an ellipse. The second option is *svd*, when two vectors x and y stay perpendicular as they travel around a circle. Then Ax and Ay move too (not usually perpendicular). The four vectors are in the Java demo on **web.mit.edu/18.06**.

The SVD is seen graphically when Ax is perpendicular to Ay. Their directions at that moment give an orthonormal basis u_1, u_2. Their lengths give the singular values σ_1, σ_2. The vectors x and y at that same moment are the orthonormal basis v_1, v_2.

Searching the Web

I will end with an application of the SVD to web search engines. When you google a word, you get a list of web sites in order of importance. You could try "four subspaces".

The HITS algorithm that we describe is one way to produce that ranked list. It begins with about 200 sites found from an index of key words, and after that we look only at links between pages. Search engines are link-based more than content-based.

Start with the 200 sites and all sites that link to them and all sites they link to. That is our list, to be put in order. Importance can be measured by links out and links in.

1. The site is an ***authority***: *links come in from many sites.* Especially from hubs.

2. The site is a ***hub***: *links go out to many sites in the list.* Especially to authorities.

We want numbers $x_1, \ldots, x_N$ to rank the authorities and $y_1, \ldots, y_N$ to rank the hubs. Start with a simple count: x_i^0 and y_i^0 count the links into and out of site i.

Here is the point: *A good authority has links from important sites* (like hubs). Links from universities count more heavily than links from friends. *A good hub is linked to important sites* (like authorities). A link to **amazon.com** unfortunately means more than a link to **wellesleycambridge.com**. The rankings x^0 and y^0 from counting links are updated to x^1 and y^1 by taking account of *good* links (measuring their quality by x^0 and y^0):

$$\textbf{Authority values} \quad x_i^1 = \sum_{j \text{ links to } i} y_j^0 \qquad \textbf{Hub values} \quad y_i^1 = \sum_{i \text{ links to } j} x_j^0 \tag{14}$$

In matrix language those are $x^1 = A^T y^0$ and $y^1 = Ax^0$. The matrix A contains 1's and 0's, with $a_{ij} = 1$ when i links to j. In the language of graphs, A is an "adjacency matrix" for the World Wide Web (an enormous matrix). The new x^1 and y^1 give better rankings, but not the best. Take another step like (14), to reach x^2 and y^2:

$$A^T A \textbf{ and } AA^T \textbf{ appear} \quad x^2 = A^T y^1 = A^T A x^0 \quad \text{and} \quad y^2 = A^T x^1 = AA^T y^0. \tag{15}$$

In two steps we are multiplying by $A^T A$ and AA^T. Twenty steps will multiply by $(A^T A)^{10}$ and $(AA^T)^{10}$. When we take powers, the largest eigenvalue σ_1^2 begins to dominate. And the vectors x and y line up with the leading eigenvectors v_1 and u_1 of $A^T A$ and AA^T. We are computing the top terms in the SVD, by the *power method* that is discussed in Section 9.3. It is wonderful that linear algebra helps to understand the Web.

Google actually creates rankings by a random walk that follows web links. The more often this random walk goes to a site, the higher the ranking. The frequency of visits gives the leading eigenvector ($\lambda = 1$) of the normalized adjacency matrix for the Web. ***That Markov matrix has 2.7 billion rows and columns, from 2.7 billion web sites.***

This is the largest eigenvalue problem ever solved. The excellent book by Langville and Meyer, *Google's PageRank and Beyond*, explains in detail the science of search engines. See `mathworks.com/company/newsletter/clevescorner/oct02_cleve.shtml`

But many of the important techniques are well-kept secrets of Google. Probably Google starts with last month's eigenvector as a first approximation, and runs the random walk very fast. To get a high ranking, you want a lot of links from important sites. The HITS algorithm is described in the 1999 *Scientific American* (June 16). But I don't think the SVD is mentioned there...

■ REVIEW OF THE KEY IDEAS ■

1. The SVD factors A into $U\Sigma V^T$, with r singular values $\sigma_1 \geq \ldots \geq \sigma_r > 0$.

2. The numbers $\sigma_1^2, \ldots, \sigma_r^2$ are the nonzero eigenvalues of AA^T and A^TA.
3. The orthonormal columns of U and V are eigenvectors of AA^T and A^TA.
4. Those columns hold orthonormal bases for the four fundamental subspaces of A.
5. Those bases diagonalize the matrix: $Av_i = \sigma_i u_i$ for $i \leq r$. This is $AV = U\Sigma$.

■ WORKED EXAMPLES ■

6.7 A Identify by name these decompositions $A = c_1b_1 + \cdots + c_rb_r$ of an m by n matrix. Each term is a rank one matrix (column c times row b). The rank of A is r.

1. *Orthogonal* columns $c_1, \ldots, c_r$ and *orthogonal* rows $b_1, \ldots, b_r$.
2. *Orthogonal* columns $c_1, \ldots, c_r$ and *triangular* rows $b_1, \ldots, b_r$.
3. *Triangular* columns $c_1, \ldots, c_r$ and *triangular* rows $b_1, \ldots, b_r$.

$A = CB$ is (m by r)(r by n). Triangular vectors c_i and b_i have zeros up to component i. The matrix C with columns c_i is lower triangular, the matrix B with rows b_i is upper triangular. Where do the rank and the pivots and singular values come into this picture?

Solution These three splittings $A = CB$ are basic to linear algebra, pure or applied:

1. **Singular Value Decomposition** $A = U\Sigma V^T$ (*orthogonal* U, *orthogonal* ΣV^T)
2. **Gram-Schmidt Orthogonalization** $A = QR$ (*orthogonal* Q, *triangular* R)
3. **Gaussian Elimination** $A = LU$ (*triangular* L, *triangular* U)

You might prefer to separate out the σ_i and pivots d_i and heights h_i:

1. $A = U\Sigma V^T$ with unit vectors in U and V. The singular values are in Σ.
2. $A = QHR$ with unit vectors in Q and diagonal 1's in R. The heights h_i are in H.
3. $A = LDU$ with diagonal 1's in L and U. The pivots are in D.

Each h_i tells the height of column i above the base from earlier columns. The volume of the full n-dimensional box ($r = m = n$) comes from $A = U\Sigma V^T = LDU = QHR$:

$$|\det A| = |\textit{product of } \sigma\text{'s}| = |\textit{product of } d\text{'s}| = |\textit{product of } h\text{'s}|.$$

6.7.B For $A = xy^T$ of rank one (2 by 2), compare $A = U\Sigma V^T$ with $A = S\Lambda S^{-1}$.

Comment This started as an exam problem in 2007. It led further and became interesting. Now there is an essay called "The Four Fundamental Subspaces: 4 Lines" on **web.mit.edu/18.06**. The Jordan form enters when $y^Tx = 0$ and $\lambda = 0$ is repeated.

6.7.C Show that $\sigma_1 \geq |\lambda|_{\max}$. The largest singular value dominates all eigenvalues. Show that $\sigma_1 \geq |a_{ij}|_{\max}$. The largest singular value dominates all entries of A.

Solution Start from $A = U\Sigma V^{\mathrm{T}}$. Remember that multiplying by an orthogonal matrix *does not change length*: $\|Qx\| = \|x\|$ because $\|Qx\|^2 = x^{\mathrm{T}}Q^{\mathrm{T}}Qx = x^{\mathrm{T}}x = \|x\|^2$. This applies to $Q = U$ and $Q = V^{\mathrm{T}}$. In between is the diagonal matrix Σ.

$$\|Ax\| = \|U\Sigma V^{\mathrm{T}}x\| = \|\Sigma V^{\mathrm{T}}x\| \leq \sigma_1\|V^{\mathrm{T}}x\| = \sigma_1\|x\|. \tag{16}$$

An eigenvector has $\|Ax\| = |\lambda|\|x\|$. So (16) says that $|\lambda|\|x\| \leq \sigma_1\|x\|$. Then $|\lambda| \leq \sigma_1$.

Apply also to the unit vector $x = (1, 0, \ldots, 0)$. Now Ax is the first column of A. Then by inequality (16), this column has length $\leq \sigma_1$. Every entry must have magnitude $\leq \sigma_1$.

Example 5 Estimate the singular values σ_1 and σ_2 of A and A^{-1}:

Eigenvalues = 1 $$A = \begin{bmatrix} 1 & 0 \\ C & 1 \end{bmatrix} \quad \text{and} \quad A^{-1} = \begin{bmatrix} 1 & 0 \\ -C & 1 \end{bmatrix}. \tag{17}$$

Solution The length of the first column is $\sqrt{1 + C^2} \leq \sigma_1$, from the reasoning above. This confirms that $\sigma_1 \geq 1$ and $\sigma_1 \geq C$. Then σ_1 dominates the eigenvalues $1, 1$ and the entry C. If C is very large then σ_1 is much bigger than the eigenvalues.

This matrix A has determinant $= 1$. $A^{\mathrm{T}}A$ also has determinant $= 1$ and then $\sigma_1\sigma_2 = 1$. For this matrix, $\sigma_1 \geq 1$ and $\sigma_1 \geq C$ lead to $\sigma_2 \leq 1$ and $\sigma_2 \leq 1/C$.

Conclusion: If $C = 1000$ then $\sigma_1 \geq 1000$ and $\sigma_2 \leq 1/1000$. *A* ***is ill-conditioned****,* slightly sick. Inverting A is easy by algebra, but solving $Ax = \boldsymbol{b}$ by elimination could be dangerous. *A is close to a singular matrix even though both eigenvalues are* $\lambda = 1$. By slightly changing the $1, 2$ entry from zero to $1/C = 1/1000$, the matrix becomes singular.

Section 9.2 will explain how the ratio $\sigma_{\max}/\sigma_{\min}$ governs the roundoff error in elimination. MATLAB warns you if this "*condition number*" is large. Here $\sigma_1/\sigma_2 \geq C^2$.

Problem Set 6.7

Problems 1–3 compute the SVD of a square singular matrix A.

1 Find the eigenvalues and unit eigenvectors v_1, v_2 of $A^{\mathrm{T}}A$. Then find $\boldsymbol{u}_1 = Av_1/\sigma_1$:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} \text{ and } A^{\mathrm{T}}A = \begin{bmatrix} 10 & 20 \\ 20 & 40 \end{bmatrix} \text{ and } AA^{\mathrm{T}} = \begin{bmatrix} 5 & 15 \\ 15 & 45 \end{bmatrix}.$$

Verify that $\boldsymbol{u}_1$ is a unit eigenvector of AA^{T}. Complete the matrices U, Σ, V.

SVD $$\begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} = \begin{bmatrix} \boldsymbol{u}_1 & \boldsymbol{u}_2 \end{bmatrix} \begin{bmatrix} \sigma_1 & \\ & 0 \end{bmatrix} \begin{bmatrix} v_1 & v_2 \end{bmatrix}^{\mathrm{T}}.$$

2 Write down orthonormal bases for the four fundamental subspaces of this A.

3 (a) Why is the trace of A^TA equal to the sum of all a_{ij}^2 ?

(b) For every rank-one matrix, why is $\sigma_1^2 =$ sum of all a_{ij}^2?

Problems 4–7 ask for the SVD of matrices of rank 2.

4 Find the eigenvalues and unit eigenvectors of A^TA and AA^T. Keep each $Av = \sigma u$:

$$\textbf{Fibonacci matrix} \qquad A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

Construct the singular value decomposition and verify that A equals $U\Sigma V^T$.

5 Use the **svd** part of the MATLAB demo **eigshow** to find those v's graphically.

6 Compute A^TA and AA^T and their eigenvalues and unit eigenvectors for V and U.

$$\textbf{Rectangular matrix} \qquad A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

Check $AV = U\Sigma$ (this will decide $\pm$ signs in U). Σ has the same shape as A.

7 What is the closest rank-one approximation to that 2 by 3 matrix?

8 A square invertible matrix has $A^{-1} = V\Sigma^{-1}U^T$. This says that the *singular values of A^{-1} are* $1/\sigma(A)$. Show that $\sigma_{\max}(A^{-1})\,\sigma_{\max}(A) \geq 1$.

9 Suppose $u_1, \ldots, u_n$ and $v_1, \ldots, v_n$ are orthonormal bases for $\mathbf{R}^n$. Construct the matrix A that transforms each v_j into u_j to give $Av_1 = u_1, \ldots, Av_n = u_n$.

10 Construct the matrix with rank one that has $Av = 12u$ for $v = \frac{1}{2}(1,1,1,1)$ and $u = \frac{1}{3}(2,2,1)$. Its only singular value is $\sigma_1 =$ _____.

11 Suppose A has orthogonal columns $w_1, w_2, \ldots, w_n$ of lengths $\sigma_1, \sigma_2, \ldots, \sigma_n$. What are U, Σ, and V in the SVD?

12 Suppose A is a 2 by 2 symmetric matrix with unit eigenvectors u_1 and u_2. If its eigenvalues are $\lambda_1 = 3$ and $\lambda_2 = -2$, what are the matrices U, Σ, V^T in its SVD?

13 If $A = QR$ with an orthogonal matrix Q, the SVD of A is almost the same as the SVD of R. Which of the three matrices U, Σ, V is changed because of Q?

14 Suppose A is invertible (with $\sigma_1 > \sigma_2 > 0$). Change A by ***as small a matrix as possible*** to produce a singular matrix A_0. Hint: U and V do not change:

$$\text{From} \qquad A = \begin{bmatrix} u_1 & u_2 \end{bmatrix} \begin{bmatrix} \sigma_1 & \\ & \sigma_2 \end{bmatrix} \begin{bmatrix} v_1 & v_2 \end{bmatrix}^T \quad \text{find the nearest } A_0.$$

15 Why doesn't the SVD for $A + I$ just use $\Sigma + I$?

Challenge Problems

16 (Search engine) Run a **random walk** $x(2), \ldots, x(n)$ starting from web site $x(1) = 1$. Count the visits to each site. At each step the code chooses the next website $x(k)$ with probabilities given by column $x(k-1)$ of A. At the end, $\boldsymbol{p}$ gives the fraction of time at each site from a histogram: count visits. ***The rankings are based on p.***

Please compare $\boldsymbol{p}$ to the steady state eigenvector of the Markov matrix A:

$$A = [0 \ .1 \ .2 \ .7; \ .05 \ 0 \ .15 \ .8; \ .15 \ .25 \ 0 \ .6; \ .1 \ .3 \ .6 \ 0]'$$

$n = 100$; $x = \text{zeros}(1, n)$; $x(1) = 1$;
for $k = 2 : n$ $x(k) = \text{min}(\text{find}(\text{rand}<\text{cumsum}(A(:, x(k-1)))))$; end
$\boldsymbol{p} = \text{hist}(x, 1 : 4)/n$

17 The $1, -1$ first difference matrix A has $A^{\mathrm{T}}A$ = second difference matrix. The singular vectors of A are *sine vectors* $\boldsymbol{v}$ and *cosine vectors* $\boldsymbol{u}$. Then $A\boldsymbol{v} = \sigma\boldsymbol{u}$ is the discrete form of $d/dx(\sin cx) = c(\cos cx)$. This is the best SVD I have seen.

SVD of A
$$A = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{bmatrix} \qquad A^{\mathrm{T}}A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

Orthogonal sine matrix
$$V = \frac{1}{\sqrt{2}} \begin{bmatrix} \sin \pi/4 & \sin 2\pi/4 & \sin 3\pi/4 \\ \sin 2\pi/4 & \sin 4\pi/4 & \sin 6\pi/4 \\ \sin 3\pi/4 & \sin 6\pi/4 & \sin 9\pi/4 \end{bmatrix}$$

(a) Put numbers in V: The unit eigenvectors of $A^{\mathrm{T}}A$ are singular vectors of A. Show that the columns of V have $A^{\mathrm{T}}A\boldsymbol{v} = \lambda\boldsymbol{v}$ with $\lambda = 2 - \sqrt{2}, \ 2, \ 2 + \sqrt{2}$.

(b) Multiply AV and verify that its columns are orthogonal. They are $\sigma_1\boldsymbol{u}_1$ and $\sigma_2\boldsymbol{u}_2$ and $\sigma_3\boldsymbol{u}_3$. The first columns of the cosine matrix U are $\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3$.

(c) Since A is 4 by 3, we need a fourth orthogonal vector $\boldsymbol{u}_4$. It comes from the nullspace of A^{T}. What is $\boldsymbol{u}_4$?

The cosine vectors in U are eigenvectors of AA^{T}. The fourth cosine is $(1, 1, 1, 1)/2$.

$$AA^{\mathrm{T}} = \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix} \qquad U = \frac{1}{\sqrt{2}} \begin{bmatrix} \cos \pi/8 & \cos 2\pi/8 & \cos 3\pi/8 \\ \cos 3\pi/8 & \cos 6\pi/8 & \cos 9\pi/8 \\ \cos 5\pi/8 & \cos 10\pi/8 & \cos 15\pi/8 \\ \cos 7\pi/8 & \cos 14\pi/8 & \cos 21\pi/8 \end{bmatrix}$$

Those angles $\pi/8, 3\pi/8, 5\pi/8, 7\pi/8$ fit 4 points with spacing $\pi/4$ between 0 and π. The sine transform has three points $\pi/4, 2\pi/4, 3\pi/4$. The full cosine transform includes $\boldsymbol{u}_4$ from the "zero frequency" or *direct current* eigenvector $(1, 1, 1, 1)$.

The 8 by 8 cosine transform in 2D is the workhorse of **jpeg compression**. Linear algebra (circulant, Toeplitz, orthogonal matrices) is at the heart of signal processing.

Table of Eigenvalues and Eigenvectors

How are the properties of a matrix reflected in its eigenvalues and eigenvectors? This question is fundamental throughout Chapter 6. A table that organizes the key facts may be helpful. Here are the special properties of the eigenvalues λ_i and the eigenvectors $\boldsymbol{x}_i$.

Symmetric: $A^{\mathrm{T}} = A$	real λ's	orthogonal $\boldsymbol{x}_i^{\mathrm{T}}\boldsymbol{x}_j = 0$
Orthogonal: $Q^{\mathrm{T}} = Q^{-1}$	all $\lvert\lambda\rvert = 1$	orthogonal $\overline{\boldsymbol{x}}_i^{\mathrm{T}}\boldsymbol{x}_j = 0$
Skew-symmetric: $A^{\mathrm{T}} = -A$	imaginary λ's	orthogonal $\overline{\boldsymbol{x}}_i^{\mathrm{T}}\boldsymbol{x}_j = 0$
Complex Hermitian: $\overline{A}^{\mathrm{T}} = A$	real λ's	orthogonal $\overline{\boldsymbol{x}}_i^{\mathrm{T}}\boldsymbol{x}_j = 0$
Positive Definite: $\boldsymbol{x}^{\mathrm{T}}A\boldsymbol{x} > 0$	all $\lambda > 0$	orthogonal since $A^{\mathrm{T}} = A$
Markov: $m_{ij} > 0, \sum_{i=1}^{n} m_{ij} = 1$	$\lambda_{\max} = 1$	steady state $\boldsymbol{x} > 0$
Similar: $B = M^{-1}AM$	$\lambda(B) = \lambda(A)$	$\boldsymbol{x}(B) = M^{-1}\boldsymbol{x}(A)$
Projection: $P = P^2 = P^{\mathrm{T}}$	$\lambda = 1;\ 0$	column space; nullspace
Plane Rotation	$e^{i\theta}$ and $e^{-i\theta}$	$\boldsymbol{x} = (1, i)$ and $(1, -i)$
Reflection: $I - 2\boldsymbol{u}\boldsymbol{u}^{\mathrm{T}}$	$\lambda = -1;\ 1, .., 1$	$\boldsymbol{u}$; whole plane $\boldsymbol{u}^{\perp}$
Rank One: $\boldsymbol{u}\boldsymbol{v}^{\mathrm{T}}$	$\lambda = \boldsymbol{v}^{\mathrm{T}}\boldsymbol{u};\ 0, .., 0$	$\boldsymbol{u}$; whole plane $\boldsymbol{v}^{\perp}$
Inverse: A^{-1}	$1/\lambda(A)$	keep eigenvectors of A
Shift: $A + cI$	$\lambda(A) + c$	keep eigenvectors of A
Stable Powers: $A^n \to 0$	all $\lvert\lambda\rvert < 1$	any eigenvectors
Stable Exponential: $e^{At} \to 0$	all Re $\lambda < 0$	any eigenvectors
Cyclic Permutation: row 1 of I last	$\lambda_k = e^{2\pi ik/n}$	$\boldsymbol{x}_k = (1, \lambda_k, \ldots, \lambda_k^{n-1})$
Tridiagonal: $-1, 2, -1$ on diagonals	$\lambda_k = 2 - 2\cos\frac{k\pi}{n+1}$	$\boldsymbol{x}_k = \left(\sin\frac{k\pi}{n+1}, \sin\frac{2k\pi}{n+1}, \ldots\right)$
Diagonalizable: $A = S\Lambda S^{-1}$	diagonal of Λ	columns of S are independent
Symmetric: $A = Q\Lambda Q^{\mathrm{T}}$	diagonal of Λ (real)	columns of Q are orthonormal
Schur: $A = QTQ^{-1}$	diagonal of T	columns of Q *if* $A^{\mathrm{T}}A = AA^{\mathrm{T}}$
Jordan: $J = M^{-1}AM$	diagonal of J	each block gives $\boldsymbol{x} = (0, .., 1, .., 0)$
Rectangular: $A = U\Sigma V^{\mathrm{T}}$	rank(A) = rank(Σ)	eigenvectors of $A^{\mathrm{T}}A$, AA^{T} in V, U